Clinical Attention Cannot Be Recovered Post Hoc: Region-Grounded Concepts for Gait-Based Assessment

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Abstract. Deep models for gait-based clinical assessment are usually explained *post hoc*, yet whether such explanations agree with what clinicians actually attend to has rarely been measured. Using two clinicians' region annotations on a gait video dataset (79 patients), we show that Grad-CAM applied to a strong 3D CNN aligns with clinician attention no better than a trivial centre prior (0.135 vs. 0.144; a no-information lower bound is 0.068). We then present a concept-grounded architecture that consumes clinician annotations *only during training* and, at inference from video alone, predicts both *which* anatomical regions a clinician would attend to and *where* they are. The predicted attention reaches an alignment of 0.537 ± 0.012, $4.0\times$ Grad-CAM, while classification remains statistically indistinguishable from the baseline (McNemar $p = 0.54$, $n = 79$ patients). Controlled ablations localise the effect: removing spatial grounding collapses alignment to the lower bound while leaving region identification intact, and replacing the grounding target with anatomically meaningless random locations—matched in shape, concentration and temporal profile—drives alignment exactly to chance (0.065). Clinical spatial attention is therefore obtainable only through explicit supervision from genuine annotations, and is not a by-product of accurate classification.

Keywords: Interpretability · Clinical priors · Gait analysis · Concept grounding.

1 Introduction

[TODO: ()]

Our contributions are:

1. A measurement showing that post-hoc explanation of an accurate gait classifier carries essentially no clinically relevant spatial information—indistinguishable from a centre prior.
2. An architecture that turns clinician attention from a model *input* into a verifiable model *output*, requiring no annotation at inference.
3. Controls establishing that the learned alignment requires genuine annotations: randomising the target location, while preserving all of its statistics, reduces alignment exactly to chance.

4. An explicit negative result: at this dataset size, classification accuracy cannot distinguish any of the configurations studied, and we report this rather than selecting a favourable seed.

2 Method

2.1 From alignment to grounding

[TODO: CLIP] Three properties of the annotations motivate the design. (i) Regions alone predict the diagnosis at exactly the majority-class rate (66.7%), so they are a *prior*, not a second modality. (ii) The corpus contains only six distinct region combinations, 60% of which are `lumbar_pelvis`, so in-batch contrastive objectives are dominated by false negatives. (iii) The two clinicians agree on only 45.7% of regions, so annotations are treated as soft targets $\{0, 0.5, 1\}$ rather than a union.

2.2 Architecture

A Kinetics-pretrained 3D ResNet-50 encodes the clip into tokens $Z \in R^{d \times T' \times H' \times W'}$. A bank of $R = 5$ clinical concepts $P \in R^{R \times d}$ queries Z by cross-attention, giving for each concept a spatio-temporal distribution A_r and a region feature $F_r = \sum_n A_r(n) Z(n)$. A shared head predicts region presence from F_r; the classifier consumes the presence-weighted concept feature concatenated with a global feature. A_r is simultaneously the pooling operator and the explanation.

2.3 Training objective

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_p \mathcal{L}_{\text{presence}} + \lambda_g \mathcal{L}_{\text{ground}} + \lambda_c \mathcal{L}_{\text{concept}}, \tag{1}$$

where $\mathcal{L}_{\text{presence}}$ is a soft BCE against the two-annotator targets and $\mathcal{L}_{\text{ground}}$ is a KL divergence between A_r and the clinician region map, weighted by the soft presence label so that unmentioned regions impose no spatial constraint. [TODO: $\mathcal{L}_{\text{concept}}$ (5.3),]

3 Experimental Setup

Data and protocol. [TODO:] 79 patients (54 ASD, 25 non-ASD). Evaluation is 5-fold cross-validation with *patient-disjoint* train/validation/test splits (≈60/20/20); validation is used only for checkpoint selection and test only for reporting. The five test folds exactly partition the 79 patients, so pooling their patient-level predictions yields 79 independent predictions—the coverage of leave-one-patient-out at one fifth of the cost.

Table 1. Alignment with clinician annotations (5 folds, mean ± s.d.). Post-hoc explanation of an accurate classifier is indistinguishable from a centre prior.

Attention source	Alignment
Ours (concept grounding)	$\mathbf{0.537 \pm 0.012}$
Grad-CAM (3D CNN baseline)	0.135 ± 0.034
Centre prior	0.144 ± 0.025
Random / uniform (no information)	0.068 ± 0.001

Table 2. Ablations (5 folds). Presence supervision determines *which* regions; spatial grounding determines *where*. The two are separable. Randomising the grounding target—preserving blob shape, concentration and temporal profile—drives alignment to the no-information bound.

Configuration	Alignment	Region AP	Macro [95% CI]
Full model	0.537 ± 0.012	0.818 ± 0.077	0.624 [0.520, 0.731]
w/o presence	0.525 ± 0.006	$\mathbf{0.468 \pm 0.114}$	0.606 [0.502, 0.714]
w/o grounding	$\mathbf{0.118 \pm 0.073}$	0.757 ± 0.093	0.654 [0.549, 0.756]
shuffled regions	0.267 ± 0.019	0.402 ± 0.034	0.627 [0.515, 0.739]
random locations	$\mathbf{0.065 \pm 0.001}$	0.770 ± 0.090	0.649 [0.531, 0.763]

Metrics. Segment-level predictions are aggregated per patient by averaging class probabilities. We report balanced accuracy (macro; majority-class baseline 0.5) with bootstrap intervals and compare configurations by exact McNemar tests on the 79 paired outcomes. Reporting at segment level ($n \approx 2800$) would overstate confidence by an order of magnitude. Alignment is score $= \sum_n A(n)\,\hat{M}(n)$ with $\hat{M} = M/\max M$, which is linear in A and therefore does not reward concentration.

Why binary. The original three-class task (ASD / DHS / LCS_HipOA) is not estimable here: LCS_HipOA has only 9 patients, leaving 1–2 per test fold under patient-disjoint splitting, and every configuration attains 0.00–0.06 recall on that class. [TODO:]

4 Results

4.1 Post-hoc explanation carries no clinical signal

4.2 Alignment requires genuine annotations

Three points form a gradient. Grounding against genuine regions gives 0.537; against a *different* genuine region (shuffled) 0.267; against a location that corresponds to no anatomy at all 0.065, exactly the no-information bound. The model therefore learns a real anatomy–concept correspondence rather than any form of spatial memorisation.

Table 3. Patient-level balanced accuracy, three seeds averaged, $n = 79$ patients. No pair of configurations differs significantly (exact McNemar).

Comparison	discordant pairs	p
3D CNN baseline vs. ours	10 : 14	0.54
w/o grounding vs. ours	8 : 6	0.79
ours vs. shuffled regions	6 : 4	0.75
ours vs. random locations	9 : 7	0.80
ours vs. no clinical prior	14 : 12	0.85

4.3 Classification is not distinguishable at this sample size

Across-seed variation for a fixed configuration reaches 0.125 in balanced accuracy, exceeding every between-configuration difference; the ranking reverses across seeds. We therefore report classification as *statistically equivalent* and make no claim of superiority. [TODO: ,]

5 Discussion

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Limitations. The cohort is small (79 patients, 25 non-ASD), which is the binding constraint on every classification comparison. Alignment is evaluated on a 7×7 token grid. The concept-contrastive term has negligible effect and is retained only for completeness. Frozen CLIP text concepts show no measurable advantage over learned ones, plausibly because the five prompts have mean pairwise cosine similarity 0.877.

6 Conclusion

Clinician spatial attention is not recoverable from an accurate classifier by post-hoc means, and is not a by-product of classification training. It can be obtained—at no measurable cost in accuracy—by supervising a concept-grounded architecture with genuine annotations, and only with genuine annotations.

References